3.1.2 Amount of substance: Sheet 8

Required practical 1: results and uncertainty · 13 marks · about 25 minutes

Key information

- Record burette readings to 2 decimal places, ending in 0 or 5 (eg 23.95 cm^3).
- Mean titre: concordant titres only (within 0.10 cm^3), given to 2 decimal places.
- **% uncertainty = uncertainty ÷ measured value × 100**. Weighing by difference uses two balance readings, so double the uncertainty.
- Total % uncertainty of an experiment = sum of the % uncertainties of each piece of apparatus.
- % purity = mass of pure substance found ÷ mass of sample used × 100.

Section A: Self-quiz

01 A volumetric flask was used to prepare 250 cm^3 of a solution. The solute was added from a plastic weighing container.

	Mass / g
Weighing container with solute	10.13
Weighing container after solute added to volumetric flask	4.48

Each reading from the balance has an uncertainty of ±0.005 g. What is the percentage uncertainty in the mass of the solute used?

[1 mark]

A 0.09%
B 0.11%
C 0.18%
D 0.22%

02 The uncertainty in 25.0 cm^3 of solution from a pipette is ±0.05 cm^3

The total uncertainty in a titre of 24.50 cm^3 of solution from a burette is ±0.15 cm^3

Calculate the total percentage error in using the pipette and burette.

[1 mark]

03 A student injects a liquid from a syringe. The mass of the syringe and contents before injecting is 11.295 g and after injecting is 10.835 g.

Each reading on the balance had an uncertainty of ±0.001 g

Calculate the percentage uncertainty in the mass of liquid injected in this experiment.

[1 mark]

04 A student places a beaker on a balance and records the mass, then adds 6 tablets and records the mass. Mass of 6 tablets = 2.14 g

Each reading from the balance has an uncertainty of ±0.005 g

Calculate the percentage uncertainty in using the balance in this experiment.

[1 mark]

05 The balance used to weigh 680 mg of iron wire has an uncertainty of ±0.005 g

A container was weighed and its mass was subtracted from the total mass of the container and wire.

Calculate the percentage uncertainty in using the balance.

[1 mark]

Section B: Exam questions

06 Citric acid, $C_3H_5O(COOH)_3$, occurs naturally in many fruits. A student analysed a sample of citric acid to determine its percentage purity.

The student dissolved 784 mg of impure citric acid in water to prepare 250 cm^3 of solution in a volumetric flask.

The student titrated 25.0 cm^3 samples of this solution with 0.0500 mol dm^{-3} sodium hydroxide solution using phenolphthalein as the indicator.

$C_3H_5O(COOH)_3(aq) + 3NaOH(aq) \rightarrow C_3H_5O(COO)_3Na_3(aq) + 3H_2O(l)$

Table 1

Titration	Rough	1	2	3
Final reading / cm^3	25.2	23.95	47.65	24.10
Start reading / cm^3	0.0	0.05	23.95	0.10
Titre / cm^3				

06.1 Complete Table 1 to show the titre in each titration.

[1 mark]

06.2 Calculate the mean titre using the concordant results.

Give your answer to the appropriate number of significant figures.

[2 marks]

06.3 The total uncertainty when using the burette is ±0.15 cm^3. This is the combination of uncertainties in the start reading, final reading and the determination of the end point.

Use your mean titre to calculate the percentage uncertainty for the use of the burette in this experiment.

[1 mark]

06.4 Use your mean titre to find the mass, in mg, of citric acid dissolved in 250 cm^3 of the solution.

The relative molecular mass (M_r) of citric acid is 192.0

[3 marks]

06.5 Calculate the percentage purity of this sample of citric acid.

[1 mark]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**C**	Source: 7404/2 Jun 2017 Q11	1
02	(0.15 ÷ 24.50 + 0.05 ÷ 25.00) × 100 = 0.8(1)%	Source: 7405/1 Jun 2023 Q06.6	1
03	% uncertainty = (2 × 0.001) ÷ 0.460 × 100 = 0.435%	Source: 7404/1 Nov 2020 Q08.3 (stem shortened)	1
04	(2 × 0.005) ÷ 2.14 × 100 = 0.467%	Source: 7404/1 Jun 2024 Q07.1 (stem shortened) 2 sf or more	1
05	1.47(%)	Source: 7405/3 Jun 2019 Q05.6 Allow 1.5(%)	1
06.1	Rough = 25.2, 1 = 23.90, 2 = 23.70, 3 = 24.00	Source: 7404/2 Jun 2018 Q02.2–02.6 Need all four (with rough to 1 dp and the other three to 2 dp)	1
06.2	M1 use of titrations 1 & 3 only M2 23.95 (cm^3)	24.0 cm^3 = 1 mark (wrong number of decimal places) 23.86 cm^3 = 1 mark (used all three titrations) Allow ECF from part 1	2
06.3	0.15 ÷ 23.95 × 100 = 0.63%	Allow any correct value with at least 2 sf based on their mean titre	1
06.4	M1 moles NaOH = 23.95 ÷ 1000 × 0.0500 (= 0.001198) M2 moles acid in flask = M1 ÷ 3 × 10 (= 0.003992) M3 mass acid (= 0.003992 × 192.0 = 0.766 g) = 766 (mg)	Correct answer to at least 2 sf = 3 marks (allow 760 – 770 mg) Correct value in grams = 2 marks Allow ECF at each stage	3
06.5	766 ÷ 784 × 100 = 97.7 or 97.8%	Allow ECF from previous part	1